

24 2025-03-27 | Week 10 | Lecture 24

24.1 Statistics: a general framework

Chapters 5.3 and 5.4

The objective of statistics is to make an inference about a population based on information contained in a sample from that population and to provide an associated measure of goodness for the inference. Here we introduce a conceptual framework for mathematical statistics.

A **population** is a large body of data that is the target of our interest. The subset collected from it is our **sample**.

Example 109 (Populations). A population can be real or theoretical. Here are some examples

- The set of people in Hawai'i (real, finite)
- The set of voters in the 2026 Midterm elections (hypothetical, finite)
- The decimal expansion of π (countably infinite)
- The infinitely many observations that could be made during a laboratory experiment if the experiment were repeated over and over again (hypothetical)
- The lifetimes of light bulbs produced by a factory

Importantly, a population can also be a *probability distribution*, specified by a pdf, pmf, or cdf

- Observations made from an exponential distribution with rate $\lambda > 0$ (i.e., the distribution with pdf $f(x) = \lambda e^{-\lambda x} \mathbf{1}_{[x>0]}$.)

End of Example 109. $\square$

More commonly in mathematical statistics, we think of the population as a distribution.

Definition 110 (Random sample from a distribution). Consider a given probability distribution on $\mathbb{R}$ that can be represented by a pdf or pmf f . We say that the random variables $X_1, \dots, X_n$ form a **random sample** from this distribution if these random variables are independent and the distribution of each is given by f . Such random variables are also said to be **independent and identically distributed** (iid). The number of random variables n is the **sample size**.

24.2 What is a statistic?

Definition 111 (Statistic). A **statistic** is any quantity whose value can be calculated from sample data.

Here are two important examples of statistics:

Example 112. Let $X_1, \dots, X_n$ be random sample.

- The **sample mean**, denoted $\bar{X}$, is

$$\bar{X} = \frac{X_1 + \dots + X_n}{n}.$$

- The **sample variance**, denoted S^2 , is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

End of Example 112. $\square$

Because a statistic is a *function* of random variables, it is itself a random variable. The probability distribution of a statistic is called its **sampling distribution**.

Theorem 113. Let $X_1, \dots, X_n$ be a random sample from a distribution with mean μ and standard deviation $\sigma > 0$. Then

- $\mu_{\bar{X}} := \mathbb{E}[\bar{X}] = \mu$
- $\sigma_{\bar{X}}^2 := \text{Var}(\bar{X}) = \sigma^2/n$ (and $\sigma_{\bar{X}} = \sigma/\sqrt{n}$)

Proof. First we compute $\mathbb{E}[\bar{X}]$:

$$\begin{aligned} \mathbb{E}[\bar{X}] &= \mathbb{E}\left[\frac{X_1 + \dots + X_n}{n}\right] && \text{by definition of } \bar{X} \\ &= \frac{1}{n} (\mathbb{E}[X_1] + \dots + \mathbb{E}[X_n]) && \text{by linearity of expectation} \\ &= \frac{1}{n} (n\mu) && \text{since } \mathbb{E}[X_i] = \mu \text{ for all } i \\ &= \mu. \end{aligned}$$

Next we compute $\text{Var}(\bar{X})$:

$$\begin{aligned} \text{Var}(\bar{X}) &= \text{Var}\left(\frac{X_1 + \dots + X_n}{n}\right) && \text{by definition of } \bar{X} \\ &= \frac{1}{n^2} \text{Var}(X_1 + \dots + X_n) && \text{by Eq. (7)} \\ &= \frac{1}{n^2} (\text{Var}(X_1) + \dots + \text{Var}(X_n)) && \text{by additivity (see Eq. (8))} \\ &= \frac{1}{n^2} (n\sigma^2) && \text{since } \text{Var}(X_i) = \sigma^2 \text{ for all } i \\ &= \frac{\sigma^2}{n}. \end{aligned}$$

□

Theorem 114. Let $X_1, \dots, X_n$ be a random sample from a normal distribution with mean μ and standard deviation σ . Then for any n , $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$.